Student Assignment Brief

This document is intended for Coventry University Group students for their own use in completing their assessed work for this module. It must not be passed to third parties or posted on any website. If you require this document in an alternative format, please contact your tutor.

Contents:

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The work you submit for this assignment must be your own independent work, or in the case of a group assignment your own groups' work. More information is available in the 'Assignment Task' section of this assignment brief.

Assignment Information

Module Name: Problem Solving and Programming

Module Code: 403IT

Assignment Title: Developing and Evaluating a Programmatic Solution to a Defined Problem

Assignment Due: Friday 12th June 2026 - 18:00 hrs

Assignment Credit: 30 Credits

Word Count (or equivalent): 1,000 – 1,500 words

Assignment Type: Composite

Percentage Grade (Applied Core Assessment). You will be provided with an overall grade between 0% and 100%. To pass the assignment you must achieve a grade of 40% or above.

Assignment Task

For this assessment, you will be presented with a defined programming scenario (see below). You must analyse the problem, plan a solution, and develop a working program using a high-level programming language. The solution should be functional, tested, and refined based on any issues encountered during development.

Alongside the code artefact, you are required to submit a reflective commentary of approximately 1,000 - 1,500 words. This written piece should explain your logical approach, design decisions, problem-solving techniques, testing processes, and how your final solution meets the original requirements.

The assessment aims to evaluate your ability to apply fundamental programming concepts, design and implement algorithms, and critically reflect on your development process.

Programming Scenario

You must choose from one of the following scenarios to complete your assessment. Each scenario requires you to design, implement, test, and reflect on a programmatic solution that meets the specified requirements. Select the one that most interests you or aligns with your current skill development goals.

Student Grade Calculator

Develop a program that allows a user to input student names and their marks across multiple modules. The program should calculate the average score for each student, assign a grade based on predefined boundaries, and generate a summary report. Your solution should include appropriate error handling for invalid inputs and allow the user to process multiple students.

Personal Budget Tracker

Create a simple personal finance tracker that enables users to record their income and expenses. The program should allow users to categorise transactions, view a running balance, and generate a monthly summary. The solution must support user-friendly data entry and include basic data validation and calculations.

Library Book Loan System

Design a program to manage the borrowing and returning of books in a small library. The system should allow users to register books, record loans and returns, check due dates, and view a list of currently borrowed books. You should incorporate basic data structures to store and manage the information and ensure that the system prevents invalid operations (e.g., borrowing a non-existent book).

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Assignment Task

1. Code Artefact

Requirements:

- Implement a solution in a high-level programming language (e.g., Python) to address the provided scenario.
- Apply appropriate problem-solving and algorithm design techniques.
- Make use of programming constructs such as variables, selection, iteration, functions/methods, and error handling where applicable.
- Test the program using appropriate data sets (if applicable) and refine the code to address any issues discovered.
- Submit fully working, well-documented code that demonstrates clear logic and structure.

Assessment Criteria:

- Clarity and correctness of the algorithm design.
- Proper use of programming syntax and control structures.
- Functionality and effectiveness of the solution.
- Quality of in-code comments and documentation.
- Evidence of testing and debugging.

2. Reflective Commentary

Word Count: 1,000 - 1,500 words

Purpose:
This section allows you to demonstrate your understanding of the development process, articulate the rationale behind your choices, and evaluate the effectiveness of your solution.

Structure:

- **Introduction**
  Briefly introduce the problem you were tasked with solving and your overall approach.
- **Logic and Planning**
  Explain the logical process you followed to analyse the problem. Include any diagrams, pseudocode, or flowcharts that supported your planning.
- **Development Process**
  Describe how you translated your plan into code. Outline key decisions, challenges you encountered, and how you addressed them.

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- **Testing and Refinement**
  Discuss your testing methods and outcomes. Highlight how you validated your code and improved it based on test results or feedback.
- **Meeting Requirements**
  Reflect on how your program aligns with the given scenario. Explain how it satisfies all functional and technical requirements.
- **Conclusion**
  Summarise what you have learned through the assessment and any skills or knowledge you have developed or strengthened.

Assessment Criteria:

- Clarity and coherence of explanation.
- Depth of analysis and critical reflection.
- Justification of development choices.
- Evidence of problem-solving and testing.
- Professional presentation, structure, and academic writing quality.

Submission Instructions:

Submission Instructions

You must submit the following:

- Fully working, well-documented code artefact.
- 1,000 – 1,500 word reflective commentary as a Word document. This must include:
  - A reference list formatted using **APA 7th edition.**
  - An **AI Use Statement**, placed in an appendix, which clearly states:
    - Which AI tools you used (e.g., ChatGPT, GitHub Copilot, Grammarly)
    - What you used them for (e.g., idea generation, code suggestions, grammar support)
    - How you verified or adapted the AI-generated content to ensure it was accurate and appropriate

⚠ *Failure to declare the use of AI tools, or misrepresenting AI-generated content as entirely your own, may result in an academic integrity investigation.*

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked the module team.

How will I receive my grades and feedback?

Provisional marks will be released once internally moderated.

Feedback will be provided by the module team alongside grades release.

Your provisional marks and feedback should be available within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found at the bottom of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align to the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task:

1. Deploy a range of problem-solving techniques.
2. Create algorithms to solve specific and generalised problems expressing them in a suitable manner to enable a programmed solution to be developed.
3. Demonstrate an understanding of the fundamentals of software development and the syntax and control structures of a high-level programming language.
4. Design, implement, test and debug programmes to solve defined problems, using appropriate tools and techniques and present them in a professional manner.

Assignment Support and Academic Integrity

If you have any questions about this assignment please see the Student Guidance on Coursework for more information.

AI Use – Expectations for This Assignment

AI Use Category: Amber

You are allowed to use AI tools to support parts of this assignment, but all submitted work must be **your own**, and **you are fully responsible** for everything you include. Misuse of AI will be treated as an **academic integrity issue**, which may result in formal investigation and penalties.

What You *May* Use AI Tools For:

AI can assist you in **supporting your learning and organisation**, but not doing the work for you. Permitted uses include:

- Helping generate **ideas, titles, structures**, or subheadings.

- **Proofreading** and checking for grammar or clarity issues.
- **Summarising** or reorganising notes or draft content.
- Creating simple **graphs, tables, or visuals** using your own data.
- **Debugging or checking errors in code** that you have written yourself.

Coding Guidance

- You **may use AI to help identify or explain issues in your code**, but:
    - The code must be written by **you**.
    - You must be able to **explain how it works**.
- You **must not** ask AI to **write full code scripts** for you - this is considered academic misconduct.

What You *Must Not* Do:

- Do **not** use AI to:
    - Write your written content for you.
    - Generate or fabricate digital evidence, logs, screenshots, or results.
    - Write or supply original code on your behalf.
    - Replace your own analysis or critical thinking.
- Do **not** upload any **personal, confidential, or sensitive information** into AI tools (some tools store and reuse your data).
- Do **not** rely on AI output without **fact-checking** using academic sources (e.g. Locate, textbooks, journals).

Spelling, Punctuation, and Grammar:

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity:

The work you submit must be your own, or in the case of groupwork, that of your group. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work. We use detection software and make routine checks for evidence of academic misconduct.

It is your responsibility to keep a record of how your thinking has developed as you progress through to submission. Appropriate evidence could include: version controlled documents, developmental sketchbooks, or journals. This evidence can be called upon if we suspect academic misconduct.

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If using Artificial Intelligence (AI) tools in the development of your assignment, you must reference which tools you have used and for what purposes you have used them. This information must be acknowledged in your final submission.

Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found on the Student Portal. All cases of suspected academic misconduct are referred for investigation, the outcomes of which can have profound consequences to your studies. For more information on academic integrity please visit the Academic and Research Integrity section of the Student Portal.

Academic Integrity – Your Responsibilities

- You are the **author** of your submission and must be able to **explain, justify, and defend** all parts of your work.
- **AI must not be used to replace your own academic thinking, decision-making, or content creation.**
- You must include a brief **AI Use Statement** at the end of your written report or within your presentation notes. This must state:
    - The name of any AI tool used (e.g. ChatGPT, Grammarly, CoPilot)
    - What it was used for (e.g. proofreading, structure suggestions)
    - How you verified and adapted its output
- You may be called for a **viva or interview** to explain how you completed your work.
- Any use of AI not declared, or any use that misrepresents AI-generated content as your own work, may be referred for an **academic integrity investigation** under the University's regulations.

If in doubt about what is allowed, **ask your module tutor** before using AI.

Support for Students with Disabilities or Additional Needs:

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms and have discussed your support needs with health and wellbeing you may be able to access support that will help with your studies.

If you feel you may benefit from additional support, but have not disclosed a disability to the University, or have disclosed but are yet to discuss your support needs it is important to let us know so we can provide the right support for your circumstances. Visit the Student Portal to find out more.

Unable to Submit on Time?

The University wants you to do your best. However, we know that sometimes events happen which mean that you cannot submit your assessment by the deadline or sit a scheduled exam. If you think

this might be the case, guidance on understanding what counts as an extenuating circumstance, and how to apply is available on the Student Portal.

Administration of Assessment

Assessment Author: [Richard Uttley]

Assessment Author Email: [ae1683@coventry.ac.uk]

Assignment Category: [Composite]

Attempt Type: [Standard]

Component Code: [CW]

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Assessment Marking Criteria

Coventry University Generic Assessment Criteria: Undergraduate

Mark band	Outcome	Guidelines
80-100% 1st	Meets learning outcomes	1st - Exceptional work with very high degree of understanding, creativity, and critical/analytic skills. Evidence of exceptional research well beyond minimum recommended using a range of methodologies. Exceptional understanding of knowledge and subject-specific theories. Demonstrates creative flair, a high degree of originality and autonomy. Exceptional ability to apply learning resources. Demonstrates well-developed problem-solving skills. Work completed with very high degree of accuracy and proficiency and autonomy. Exceptional communication and expression, significant evidence of professional skill set. Student evidences deployment of a full range of exceptional technical, including proficiency in the English Language, and/or artistic skills.
70-79% 1st		1st - Excellent work with clear evidence of understanding, creativity and critical/analytical skills. Thorough research well beyond the minimum recommended using methodologies beyond the usual range. Excellent understanding of knowledge and subject-specific theories with evidence of considerable originality and autonomy. Excellent ability to apply learning resources. Demonstrates consistent, coherent substantiated argument and interpretation. Demonstrates considerable creativity and clear problem-solving skills. Assessment completed with accuracy, proficiency, and considerable autonomy. Excellent communication and expression, some evidence of professional skill set. Student evidences deployment of a highly developed range of technical, including proficiency in the English Language, and/or artistic skills.
60-69% 2:1		2:1 - Very good work demonstrating strong understanding of theories, concepts and issues with clear critical analysis. Thorough research, using established methodologies accurately, beyond the recommended minimum with little, if any, irrelevant material present. Very good understanding, evidencing breadth and depth, of knowledge and subject-specific theories with some originality and autonomy. Very good ability to apply learning resources. Demonstrates coherent substantiated argument and interpretation. Demonstrates some originality, creativity and problem-solving skills. Work completed with accuracy, proficiency, and autonomy. Very good communication and expression with evidence of professional skill set. Student has a thorough command of a good range of technical, including proficiency in the English Language, and/or artistic skills.

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50-59% 2:2		2:2 - Good understanding of relevant theories, concepts and issues with some critical analysis. Research undertaken accurately using established methodologies, enquiry beyond that recommended may be present. Some errors may be present and some inclusion of irrelevant material. Good understanding, with evidence of breadth and depth, of knowledge and subject-specific theories with indications of originality and autonomy. Good ability to apply learning resources. Demonstrates logical argument and interpretation with supporting evidence. Demonstrates some originality, creativity and problem-solving skills but with inconsistencies. Expression and presentation mostly accurate, proficient, and conducted with some autonomy. Good communication and expression with appropriate professional skill set. Student consistently demonstrates a well-developed range of technical, including proficiency in the English Language, and/or artistic skills.
40-49% 3rd Class		3rd - Meet the learning outcomes with a basic understanding of relevant theories, concepts and issues. Demonstrates an understanding of knowledge and subject-specific theories sufficient to deal with concepts. Assessment may be incomplete and with some errors. Research scope sufficient to evidence use of some established methodologies. Some irrelevant material likely to be present. Basic ability to apply learning resources. Demonstrates ability to devise and sustain an argument. Demonstrates some originality, creativity and problem-solving skills but with inconsistencies. Expression and presentation sufficient for accuracy and proficiency. Sufficient communication and expression with basic professional skill set. Student demonstrates technical, including a basic level of proficiency in the English Language, and/or artistic skills.
30-39% Fail	Fails to achieve learning outcomes	Fail – Outcomes not met. Limited understanding of relevant theories, concepts and issues. Little evidence of research and use of established methodologies. Some relevant material will be present. Deficiencies evident in analysis. Fundamental errors and some misunderstanding likely to be present. Limited ability to apply learning resources. Student's arguments are weak and poorly constructed. Limited originality, creativity, and struggles with problem-solving skills. Expression and presentation insufficient for accuracy and proficiency. Insufficient communication and expression and with deficiencies in professional skill set. Student demonstrates some deficiencies in technical, including in their use of the English Language, and/or artistic skills.
0-29% Fail		Fail – Outcomes not met. Clear failure demonstrating very little understanding of relevant theories, concepts and issues. Minimal evidence of research and use of established methodologies and incomplete knowledge of the area. Serious and fundamental errors and aspects missing. Very little evidence of ability to apply learning resources. Student's arguments are very weak and with no evidence of alternative views. Little evidence of originality, creativity, and problem-solving skills. Expression and presentation

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		deficient for accuracy and proficiency. Insufficient communication and expression and with deficiencies in professional skill set. Student demonstrates a lack of technical, including in their use of the English Language, and/or artistic skills.

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